

Linear Regression

Linear regression is a **supervised learning algorithm** used for predicting a **continuous output** based on an input variable(s). It establishes a relationship between **independent variable(s) (X)** and a **dependent variable (Y)** by fitting a **straight line** to the data.

Bivariate Data in Linear Regression

Bivariate data consists of **two variables** that are related to each other. In **linear regression**, these two variables are typically classified as:

1. **Independent Variable (X)** – The predictor or explanatory variable.
2. **Dependent Variable (Y)** – The response or outcome variable.

Best-Fitted Line in Linear Regression

The **best-fitted line** (also called the **regression line**) is the straight line that **best represents the relationship** between the independent variable X and the dependent variable Y . It minimizes the **error (residuals)** between the actual data points and the predicted values.

Mathematical Representation

The best-fitted line follows the equation:

$$Y = mX + c$$

where:

- Y = Predicted dependent variable.
- X = Independent variable.
- m = Slope (rate of change of Y with respect to X).
- c = Intercept (the value of Y when $X = 0$).

11.1 A study was conducted at Virginia Tech to determine if certain static arm-strength measures have an influence on the “dynamic lift” characteristics of an individual. Twenty-five individuals were subjected to strength tests and then were asked to perform a weight-lifting test in which weight was dynamically lifted overhead. The data are given here.

Individual	Arm Strength, x	Dynamic Lift, y
1	17.3	71.7
2	19.3	48.3
3	19.5	88.3
4	19.7	75.0
5	22.9	91.7
6	23.1	100.0
7	26.4	73.3
8	26.8	65.0
9	27.6	75.0
10	28.1	88.3
11	28.2	68.3
12	28.7	96.7
13	29.0	76.7
14	29.6	78.3
15	29.9	60.0
16	29.9	71.7
17	30.3	85.0
18	31.3	85.0
19	36.0	88.3
20	39.5	100.0
21	40.4	100.0
22	44.3	100.0
23	44.6	91.7
24	50.4	100.0
25	55.9	71.7

- (a) Estimate β_0 and β_1 for the linear regression curve $\mu_{Y|x} = \beta_0 + \beta_1 x$.
- (b) Find a point estimate of $\mu_{Y|30}$.
- (c) Plot the residuals versus the x 's (arm strength). Comment

(a) Estimate β_0 and β_1 for the linear regression curve $\mu_{Y|x} = \beta_0 + \beta_1 x$

The regression line is of the form:

$$y = \beta_1 x + \beta_0$$

where:

- β_1 is the slope,
- β_0 is the y-intercept.

Step 1: Calculate the necessary sums

Given:

- $\sum x_i = 778.7$
- $\sum y_i = 2050.0$
- $\sum x_i^2 = 26,591.63$
- $\sum x_i y_i = 65,164.04$
- $n = 25$

Step 2: Calculate the slope (β_1) and y-intercept (β_0)

The formulas for the slope (β_1) and y-intercept (β_0) are:

$$\beta_1 = \frac{n(\sum x_i y_i) - (\sum x_i)(\sum y_i)}{n(\sum x_i^2) - (\sum x_i)^2}$$
$$\beta_0 = \frac{\sum y_i - \beta_1(\sum x_i)}{n}$$

Calculations:

- $\beta_1 = \frac{(25)(65,164.04) - (778.7)(2050.0)}{(25)(26,591.63) - (778.7)^2} = 0.5609$
- $\beta_0 = \frac{2050.0 - (0.5609)(778.7)}{25} = 64.53$

(b) Find a point estimate of $\mu_{Y|30}$

To estimate the dynamic lift ($\hat{y}$) when arm strength (x) is 30, use the regression equation:

$$\hat{y} = 0.5609x + 64.53$$

Substitute $x = 30$:

$$\hat{y} = 0.5609(30) + 64.53$$

$$\hat{y} = 16.827 + 64.53$$

$$\hat{y} = 81.357 \approx 81.36$$

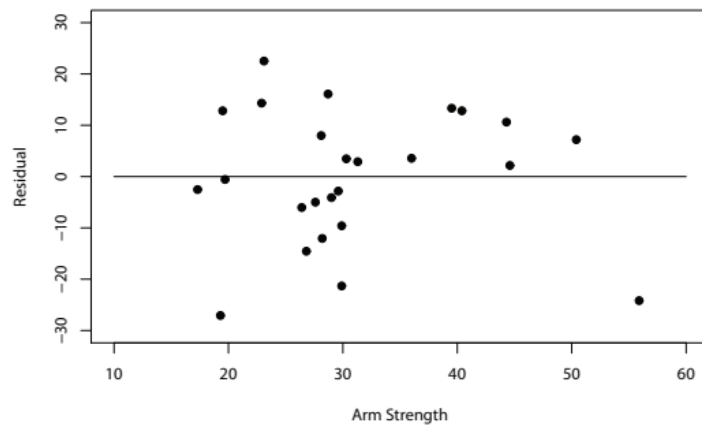
Final Answers

(a) The regression equation is:

$$y = 0.5609x + 64.53$$

(b) The point estimate of $\mu_{Y|30}$ is approximately **81.36**.

(c) Residuals appear to be random as desired.



11.2 The grades of a class of 9 students on a midterm report (x) and on the final examination (y) are as follows:

x	77	50	71	72	81	94	96	99	67
y	82	66	78	34	47	85	99	99	68

- Estimate the linear regression line.
- Estimate the final examination grade of a student who received a grade of 85 on the midterm report.

(a) Estimating the Linear Regression Line

The equation of a straight line is:

$$y = mx + c$$

where:

- m is the slope,
- c is the intercept.

Given the data:

- $\sum x_i = 707$,
- $\sum y_i = 658$,
- $\sum x_i^2 = 57,557$,
- $\sum x_i y_i = 53,258$,
- $n = 9$,

The slope m is:

$$m = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)^2}$$
$$m = \frac{(9)(53,258) - (707)(658)}{(9)(57,557) - (707)^2}$$
$$m = 0.7771$$

The intercept c is:

$$c = \frac{\sum y_i - m \sum x_i}{n}$$
$$c = \frac{658 - (0.7771)(707)}{9}$$
$$c = 12.0623$$

Thus, the estimated regression equation is:

$$y = 0.7771x + 12.0623$$

(b) Predicting the Final Exam Grade for $x = 85$

Using the equation:

$$y = 0.7771(85) + 12.0623$$
$$y = 66.0535 + 12.0623$$
$$y = 78$$

Thus, for a midterm score of 85, the estimated final exam grade is 78.

11.3 The amounts of a chemical compound y that dissolved in 100 grams of water at various temperatures x were recorded as follows:

x ($^{\circ}\text{C}$)	y (grams)		
0	8	6	8
15	12	10	14
30	25	21	24
45	31	33	28
60	44	39	42
75	48	51	44

- Find the equation of the regression line.
- Graph the line on a scatter diagram.
- Estimate the amount of chemical that will dissolve in 100 grams of water at 50°C .

(a) Finding the Regression Line

The equation of a straight line is:

$$y = mx + c$$

where:

- m is the slope,
- c is the intercept.

Given the summations:

- $\sum x_i = 675$
- $\sum y_i = 488$
- $\sum x_i^2 = 37,125$
- $\sum x_i y_i = 25,005$
- $n = 18$

Step 1: Calculate the Slope m

$$m = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)^2}$$

$$m = \frac{(18)(25,005) - (675)(488)}{(18)(37,125) - (675)^2}$$

$$m = 0.5676$$

Step 2: Calculate the Intercept c

$$c = \frac{\sum y_i - m \sum x_i}{n}$$

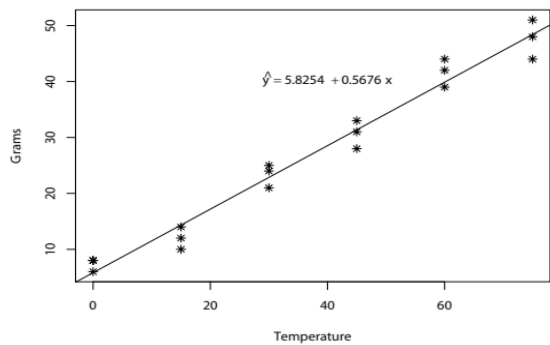
$$c = \frac{488 - (0.5676)(675)}{18}$$

$$c = 5.8254$$

Thus, the regression equation is:

$$y = 0.5676x + 5.8254$$

(b) The scatter plot and the regression line are shown below.



(c) Estimate the Amount of Chemical at $x = 50^{\circ}C$

Using the regression equation:

$$y = 0.5676(50) + 5.8254$$

$$y = 28.38 + 5.8254$$

$$y = 34.205 \text{ grams}$$

Thus, at $50^{\circ}C$, the estimated amount of chemical that will dissolve in 100 grams of water is **34.205 grams**.

11.4 The following data were collected to determine the relationship between pressure and the corresponding scale reading for the purpose of calibration.

Pressure, x (lb/sq in.)	Scale Reading, y
10	13
10	18
10	16
10	15
10	20
50	86
50	90
50	88
50	88
50	92

- Find the equation of the regression line.
- The purpose of calibration in this application is to estimate pressure from an observed scale reading. Estimate the pressure for a scale reading of 54 using $\hat{x} = (54 - b_0)/b_1$.

(a) Estimating the Linear Regression Line

The equation of a straight line is:

$$y = mx + c$$

where:

- m is the slope,
- c is the intercept.

Given the summations:

- $\sum x_i = 16.5$
- $\sum y_i = 100.4$
- $\sum x_i^2 = 25.85$
- $\sum x_i y_i = 152.59$
- $n = 11$

Step 1: Calculate the Slope m

$$m = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)^2}$$
$$m = \frac{(11)(152.59) - (16.5)(100.4)}{(11)(25.85) - (16.5)^2}$$
$$m = 1.8091$$

Step 2: Calculate the Intercept c

$$c = \frac{\sum y_i - m \sum x_i}{n}$$
$$c = \frac{100.4 - (1.8091)(16.5)}{11}$$
$$c = 6.4136$$

Thus, the estimated regression equation is:

$$y = 1.8091x + 6.4136$$

(b) Estimate the pressure for a scale reading of 54

To estimate the pressure ($\hat{x}$) for a scale reading of 54, use the formula:

$$\hat{x} = \frac{54 - b_0}{b_1}$$

Substitute the values of b_0 and b_1 :

$$\hat{x} = \frac{54 - (-1.7)}{1.81} = \frac{54 + 1.7}{1.81} = \frac{55.7}{1.81} \approx 30.78$$

Final Answers

(a) The equation of the regression line is:

$$y = 1.81x - 1.7$$

(b) The estimated pressure for a scale reading of 54 is approximately **30.78 lb/sq in.**

11.5 A study was made on the amount of converted sugar in a certain process at various temperatures. The data were coded and recorded as follows:

Temperature, x	Converted Sugar, y
1.0	8.1
1.1	7.8
1.2	8.5
1.3	9.8
1.4	9.5
1.5	8.9
1.6	8.6
1.7	10.2
1.8	9.3
1.9	9.2
2.0	10.5

- (a) Estimate the linear regression line.
- (b) Estimate the mean amount of converted sugar produced when the coded temperature is 1.75.
- (c) Plot the residuals versus temperature. Comment.

(a) Estimating the Linear Regression Line

The equation of a straight line is:

$$y = mx + c$$

where:

- m is the slope,
- c is the intercept.

Given the summations:

- $\sum x_i = 16.5$
- $\sum y_i = 100.4$
- $\sum x_i^2 = 25.85$
- $\sum x_i y_i = 152.59$
- $n = 11$

Step 1: Calculate the Slope m

$$m = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)^2}$$
$$m = \frac{(11)(152.59) - (16.5)(100.4)}{(11)(25.85) - (16.5)^2}$$
$$m = 1.8091$$

Step 2: Calculate the Intercept c

$$c = \frac{\sum y_i - m \sum x_i}{n}$$
$$c = \frac{100.4 - (1.8091)(16.5)}{11}$$
$$c = 6.4136$$

Thus, the estimated regression equation is:

$$y = 1.8091x + 6.4136$$

(b) Estimate the Mean Amount of Converted Sugar for $x = 1.75$

Using the regression equation:

$$y = 1.8091(1.75) + 6.4136$$
$$y = 3.1660 + 6.4136$$
$$y = 9.580$$

Thus, for $x = 1.75$, the estimated amount of converted sugar is **9.580**.

c) Residuals appear to be random as desired.

